Atmo SHEILD

AI Air Pollution & Health Advisory System

"Breathe smarter. Understand your air."

1. Executive Summary & Vision

Atmo SHEILD is an intelligent environmental telemetry platform engineered to translate raw atmospheric data into understandable, proactive human health actions. Designed to address the global crisis of airborne particulate exposure, the application combines open meteorological models, physical IoT sensor detectors, an intuitive 3-tier circular AQI meter, neighborhood station mapping, and automated AI health advisories customized across 7 distinct physiological vulnerability cohorts.

THE PROBLEM

• Ambient particulate matter (PM2.5, PM10) causes millions of premature pulmonary and cardiovascular complications annually.

• Standard weather services display isolated numerical indices (e.g., "AQI 142") with zero health translation.

• High-risk groups (children, seniors, asthmatics) lack actionable, physiology-specific precautions.

THE SOLUTION

• Universal 3-tier classification: LOW (0-50), MEDIUM (51-100), and HIGH (>100) with clear layman definitions.

• Contextual AI health advisory tailored to 7 sensitivity groups (General, Kids, Seniors, Respiratory, Athletes, Workers, Farming).

• Dual telemetry architecture bridging physical IoT laser sensors and global atmospheric model APIs.

2. Core Platform Capabilities

Real-Time AQI	Local Stations	3-Tier Gauge	AI Advisory	Air History
7 Pollutants & Weather telemetry	A-Z Search & GPS Leaflet map	Circular SVG color-sweep meter	7 physiological precaution groups	Weekly & Monthly verified trends

Coding Languages & Technologies Used

Atmo SHEILD implements a production-grade decoupled architecture with a Python backend and TypeScript React frontend.

PYTHON (Python 3.10)

Backend Core & REST API

- Used for core logic in app.py and services (aqi_service.py, weather_service.py, location_service.py, health_advisory.py, history_service.py, database.py).
- Flask framework provides REST API endpoints (/api/aqi, /api/nearby, /api/advisory, /api/history).
- Handles sensor packet processing, Haversine spherical distance calculations, and rolling statistical models.

SQL (SQLite3)

Database & Relational Persistence

- ACID-compliant relational database located at database/aqi_history.db.
- Structured tables: aqi_readings (telemetry), locations (geo-coordinates), and user_profiles.
- Powers 30-day historical time-series analytics and period-over-period trend validation queries.

TYPESCRIPT (TypeScript 5.8)

Type-Safe Frontend Interface

- Component development with strict interfaces (AirQualityData, LocationItem, HealthAdvisory, MonthlyHistory).
- Eliminates runtime null reference errors during environmental sensor and telemetry parsing.
- React 19 hooks manage state synchronization across maps, gauges, and dynamic health advisory selectors.

JAVASCRIPT (ES2022)

Client Execution & Map Drivers

- Executes browser Geolocation API handlers and asynchronous fetch pipelines.
- Drives Leaflet.js interactive OpenStreetMap tile caching and marker events.
- Standalone static script bundled in static/js/script.js for direct Jinja view execution.

HTML5 & CSS3 / TAILWIND CSS v4

Accessible Markup & Modern Design System

- Semantic HTML5 structure with W3C web accessibility guidelines and OpenGraph metadata.
- Tailwind CSS v4 utility architecture featuring soft glassmorphism, responsive grids, and zero-pill discipline.
- Dynamic CSS animations for the 240-degree SVG circular AQI meter and subtle atmospheric flow effects.

Live AQI, Weather & 3-Tier Classification

Atmo SHEILD integrates dual telemetry feeds with an intuitive 3-tier visual gauge and zero-synthetic data policy.

Ø=ßâ CLEAN AIR

LOW (0 - 50 AQI)

Air quality is relatively good. Atmospheric particulates are well within safe thresholds. Optimal for all school recess, senior morning walks, and marathon training.

Ø=ßá MODERATE

MEDIUM (51 - 100 AQI)

Moderate pollution. May affect sensitive individuals (asthma, COPD, children) during prolonged outdoor exposure. Consider pacing cardio workouts away from heavy traffic.

Ø=Ý4 ELEVATED RISK

HIGH (> 100 AQI)

Significant particulate burden. Everyone may experience adverse health effects. Sensitive cohorts should stay indoors with HEPA filtration; wear N95 respirators if outside.

Monitored Environmental Parameters

PM2.5
Fine respirable soot "d 2

PM10
Inhalable coarse dust "d

CO
Carbon Monoxide (mg/m³)

NO2
Nitrogen Dioxide from traffic

SO2
Sulfur Dioxide from industry

O3
Ground-level Photochemical Ozone

Temperature
Ambient thermal gradient (°C)

Humidity & Wind
Atmospheric dispersion factor

Strict Zero-Synthetic Data Protocol

In strict compliance with environmental health reliability standards, if live IoT sensor telemetry or atmospheric station endpoints are unreachable, Atmo SHEILD displays "Data unavailable" with clear diagnostics instead of fabricating simulated numbers.

Omni-Search Engine & Interactive Station Map

Combines real-time autocompletion, global geocoding, A-Z filtering, and Leaflet OpenStreetMap telemetry.

1. Debounced Omni-Search Bar (Fixed & Enhanced)

Users can search local pre-loaded stations (Master Canteen, Janpath Road, Kalpana Square, Khandagiri, Patia) as well as any city globally (Paris, Tokyo, New York, Delhi). Features live dropdown suggestions with distance tags, clear button, and keyboard Enter/Escape support.

2. Strict Alphabetical Order (A-Z) Dropdown

Synchronized select element sorting all monitoring hubs alphabetically from Bengaluru to Tokyo, enabling rapid non-typing navigation for mobile and tablet users.

3. Browser GPS Geolocation with Haversine Math

One-click position detection calculates spherical kilometer distance to all regional stations using the Haversine equation: d = 2R · atan2(" a, " (1-a)), dynamically snapping the user to their closest monitoring sensor.

4. Interactive Leaflet OpenStreetMap Layer

Dynamic interactive map rendering color-coded live AQI pins (Ø=ßâ Low, Ø=ßá Med, Ø=Ý4 High). immediately refreshes the active dashboard, weather parameters, and AI health advisory.

Sample Regional Stations Telemetry (Bhubaneswar Hub)

Station Name	Area / Sector	Coordinates	AQI Index	Level Status
Master Canteen	Station Square, Bhubaneswar	20.2644° N, 85.8394° E	85 AQI	M E D I U M Ø=ßá
Janpath Road	Commercial Corridor	20.2818° N, 85.8436° E	78 AQI	M E D I U M Ø=ßá
Kalpana Square	South Junction	20.2520° N, 85.8378° E	110 AQI	H I G H Ø=Ý4
Patia IT Corridor	North Tech Zone	20.3588° N, 85.8164° E	48 AQI	L O W Ø=ßâ

Contextual AI Health Advisory System

Evaluates real-time pollutants and meteorological variables to generate customized guidance across 7 cohorts.

7 Physiological Sensitivity Cohorts

• **General Public:**	Baseline cellular health; standard hydration & ventilation routines.
• **Children:**	Developing alveoli; higher respiratory air intake per kg body weight.
• **Elderly (65+):**	Aging cardiovascular defenses; particulate stroke & ischemic risks.
• **Respiratory-Sensitive:**	Asthma, COPD, allergies; prevention of acute bronchospasms.
• **Outdoor / Sports:**	High-ventilation endurance training; deep alveolar soot mitigation.
• **Outdoor Workers:**	Construction & traffic crews; extended shifts & N95 shift rotation.
• **Farming & Pets:**	Trough water cleanliness, livestock plume shelter, domestic pets.

WHAT TO DO

• Stay hydrated to keep respiratory mucus membranes moist and resilient.

• Wear snug-fitted N95 / KN95 particulate respirators when AQI exceeds 100.

• Seal indoor windows and operate HEPA purifiers during peak morning traffic.

• Check Atmo SHEILD hourly trend curves before planning outdoor departures.

• Move cardio training indoors when elevated particulates are detected.

WHAT NOT TO DO

• Avoid unnecessary prolonged outdoor exposure during high pollution.

• Avoid strenuous aerobic workouts along arterial highway corridors.

• Avoid burning trash, leaves, or agricultural biomass.

• Avoid unventilated indoor smoking, frying, or heavy incense burning.

• Never ignore severe municipal air pollution alerts or symptoms.

Medical Disclaimer: Atmo SHEILD provides general environmental wellness guidance based on WHO/EPA ambient standards and does not diagnose, treat, or replace

Weekly & Monthly Pollution Trend Engine

Time-series analysis using SQLite database logs, rolling arithmetic averages, and verified comparisons.

WEEKLY RECAP (7-DAY ROLLING METRICS)

Average AQI	Lowest Day	Highest Peak	Category Count	Average PM2.5
82 AQI	**45 AQI**	**142 AQI**	**2 Low / 4 Med / 1 High**	**34.4 µg/m³**
7-day mean	Pruned clean air	Episode spike	Days distribution	Ambient burden

MONTHLY RECAP (30-DAY OBSERVATIONS)

Monthly Mean	Lowest Day	Highest Day	High-AQI Days	Trend Direction
86 AQI	**48 AQI**	**114 AQI**	**3 Days**	**Improved**
September average	September 15	September 20	Exceeded AQI 100	-6 pts vs August

Mathematical Verification Guarantee (Zero Hallucination)

Statements such as "AQI improved compared with last month" are strictly governed by an automated regression comparator in history_service.py. If historical rows stored in the SQLite database do not corroborate a lower period-over-period mean by at least 2 AQI points, the system displays neutral consistency rather than false improvements.

Smart Pollution Alert Trigger Engine

When local telemetry crosses 100 AQI, an active high-alert notification banner triggers across the UI with audible bells and protective guidance. Users can

Relational Database Schema & API Contracts

Clean modular design ensuring scale-to-zero efficiency, low latency (<150ms), and zero external dependencies.

SQLITE DATABASE SCHEMA: database/aqi_history.db

```
CREATE TABLE aqi_readings (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  location_name TEXT NOT NULL,
  aqi INTEGER NOT NULL, pm2_5 REAL, pm10 REAL,
  co REAL, no2 REAL, so2 REAL, o3 REAL,
  temperature REAL, humidity REAL, wind_speed REAL,
  weather_condition TEXT, aqi_level TEXT NOT NULL,
  source_type TEXT DEFAULT "api",
  recorded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
CREATE TABLE locations (id INTEGER PRIMARY KEY, name TEXT UNIQUE, city TEXT, lat REAL, lon REAL);
CREATE TABLE user_profiles (id INTEGER PRIMARY KEY, username TEXT, sensitivity_group TEXT, alerts INTEGER);
```

REST API Specification

Method	Endpoint	Query Parameters	Output Payload
GET	/api/aqi	location, lat, lon, source	AQI, 7 pollutants, weather, status
GET	/api/nearby	lat, lon, limit	Array of regional stations & AQIs
POST	/api/advisory	aqi, pm2_5, group, weather	Contextual AI advisory & action lists
GET	/api/history/weekly	location	7-day rolling statistics & line points
GET	/api/history/monthly	location	30-day stats & verified trend statement
GET	/api/locations	q (query)	Alphabetically sorted location list

Production Roadmap:

- Phase 1 (Completed): Full-stack Flask REST API, SQLite persistence, React 19 UI, and 3-tier circular gauge.
- Phase 2 (Near-Term): Direct MQTT websocket integration for physical ESP32 / Arduino PMS5003 WiFi sensor nodes.
- Phase 3 (Scale): Progressive Web App (PWA) offline caching and municipal emergency broadcast webhooks.